AI-assisted glioma research: a perspective with reproducible examples in imaging and clinical evidence

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Abstract

Glioma research must connect anatomical tumour burden with the cells, circuits and networks that occupy the same brain, and with the cognition of the patient. This perspective is organised around twelve executable notebooks, each pairing one clinical or biological question with code, controls and an explicit evidence label. The examples run from image to cognition: measuring tumour compartments and recomputing a published segmentation cohort; following one openly released glioblastoma over serial scans with a growth model whose failure is diagnosed and revised; constructed models of growth, cell-state composition, neuron–glioma feedback, astrocytic potassium clearance and extracellular fields; a synthetic network lesioned by a tissue-guided tumour; one patient's pre- and post-operative connectomes compared with a virtual resection; and cognitive test scores from an open brain-tumour cohort, including an audit of an AI-generated summary slide. Three kinds of AI are kept apart: trained imaging models, generative research assistants that helped write the code and text, and conventional numerical solvers that reproduce specified equations without a language-model call. A final example shows how records and independent checks make an assisted contribution inspectable, and why passing checks is not evidence of scientific quality or benefit. A thirteenth notebook, a guide without new results, places AI assistance in each of nine stages of a research workflow, from idea to further studies. Synthetic constructions, real-data observations, published results, hypotheses and proposed studies are labelled separately. No model here is calibrated to a patient, no clinical validation is claimed, and the comparative benefit of AI assistance is not measured.

Keywords: glioma; MRI; tumour growth models; cancer neuroscience; connectome; cognition; AI assistance; reproducibility

1 Introduction

The value of a computational glioma figure depends on the question it answers. A segmentation may define a region for measurement, an equation may construct a trajectory, a connectome may suggest which connections a tumour interrupts, and a cognitive table may describe how a patient performed. Each can be useful without supporting a treatment conclusion.

Three kinds of AI and computation appear in this work and are kept apart. An image-analysis model, such as one developed with nnU-Net, maps an image to labels and requires validation of that mapping [1]. A generative research assistant can help formulate questions, write code, inspect artifacts and revise prose. A numerical solver advances a specified equation. The solver is not an imaging model, and the correctness of its update does not validate the assistant that helped implement it. Rebuilding every figure in this paper runs conventional code only and calls no language model.

The paper accompanies a public repository of thirteen notebooks (Table 1). Each notebook asks one question, calls a shared Python package, and states what kind of evidence it supplies.

We use five evidence labels throughout: SYNTHETIC (a construction whose inputs are prescribed), PUBLISHED (a result reported by others, quoted or recomputed from released files), OBSERVATION (real data, described without a fitted model or causal claim), HYPOTHESIS (a proposed mechanism not tested here) and PROPOSED (a study design, not carried out). Every section below opens with a short block that states the question, the AI contribution, the investigator's choices, the verification and the evidence status.

The sections follow a path from image to cognition. Section 2 measures and locates a tumour; Section 3 follows one patient's serial scans; Section 4 turns to growth and its cellular environment; Section 5 to physiology at the glioma–brain interface; Section 6 to networks and cognition; and Section 7 to auditing the AI-assisted work itself.

Table 1: The thirteen notebooks, the question each asks, its evidence label and where it appears in this paper. Notebook 00 is a Colab quickstart and is not part of the scientific content; notebook 13 is a guide that adds no result.

No.	Question	Evidence	In this paper
01	How do overlap and volume respond to segmentation errors, and how does a published model agree with reference masks?	synthetic; published	Sec. 2.1, Fig. 1, 2
02	How do migration, proliferation and a display threshold change a simulated tumour?	synthetic	Sec. 4.1, Fig. 7
03	Can a growth model started from one visit reproduce the automated labels four weeks later?	observation; synthetic	Sec. 3, Fig. 5, 6
04	Which records and checks make AI-assisted code inspectable?	synthetic; records	Sec. 7, Tab. 3, 4
05	Can tumour composition change at fixed abundance?	synthetic	Sec. 4.2, Fig. 8
06	What separates one-way growth modulation from neuron–glioma feedback?	synthetic	Sec. 4.3, Fig. 9
07	What changes when astrocytic potassium clearance slows?	synthetic	Sec. 5.1, Fig. 10
08	Does a recorded potential identify a field or a cognitive mechanism?	synthetic; hypothesis	Sec. 5.2, Fig. 11
09	What does a published segmentation study measure, and where does the tumour end?	published	Sec. 2.2, Tab. 2, Fig. 3
10	What did patients score before surgery and at follow-up, and when is a change real?	observation	Sec. 6.3, Fig. 15
11	How does a tissue-guided tumour and its resection change a network?	synthetic	Sec. 6.1, Fig. 12
12	Does a virtual resection predict one patient's post-operative network?	observation	Sec. 6.2, Fig. 13, 14
13	Where can AI assistance enter each stage of a research project, and what must the researcher still check?	published; records; proposed	Sec. 8

The clinical motivation includes preservation of language and other cognitive functions alongside control of tumour burden. Anatomical proximity, physiological recruitment, network disconnection and task performance are distinct observations. Their integration is a research question, not a consequence of segmentation accuracy.

2 Measuring and locating a tumour

2.1 Constructed compartments (notebook 01)

Question: How do geometry and mask perturbations change measurement?
AI contribution: Research assistance adapting unit-explicit measurement code and teaching controls.
Investigator choices: Imaging was requested; grid, labels and perturbations are prescribed construction choices.
Verification: Affine volume, Dice, matched grids and refusal of ambiguous geometry.
Evidence status: SYNTHETIC; independent of any trained imaging model.

The construction is an ellipsoid on a $1.2 \times 1.2 \times 2.0$ mm grid with nested compartments in BraTS label numbering: 1 for necrosis or non-enhancing tumour, 2 for oedema or FLAIR abnormality and 4 for enhancing tumour, arranged as in a typical glioblastoma. These are prescribed regions, not observed tissue. For a binary region M and a voxel-to-world affine A in millimetres,

$$V(M) = \frac{|M|\,|\det A_{1:3,1:3}|}{1000} \quad \text{mL}. \tag{1}$$

The determinant handles rotation, reflection and shear. The NIfTI reader requires a coded transform, established units and finite integer labels, and rejects conflicting qform and sform transforms. Comparisons require the same physical grid; the code never registers or resamples silently.

A four-voxel translation is compared with a 12% increase in each semi-axis. The reference whole tumour measures 8.018 mL. The translated mask keeps that volume but has Dice 0.644; the expanded mask has Dice 0.830 and 41.09% more volume (Figure 1). Overlap and volume therefore answer different questions, and a high Dice score does not guarantee an accurate volume.

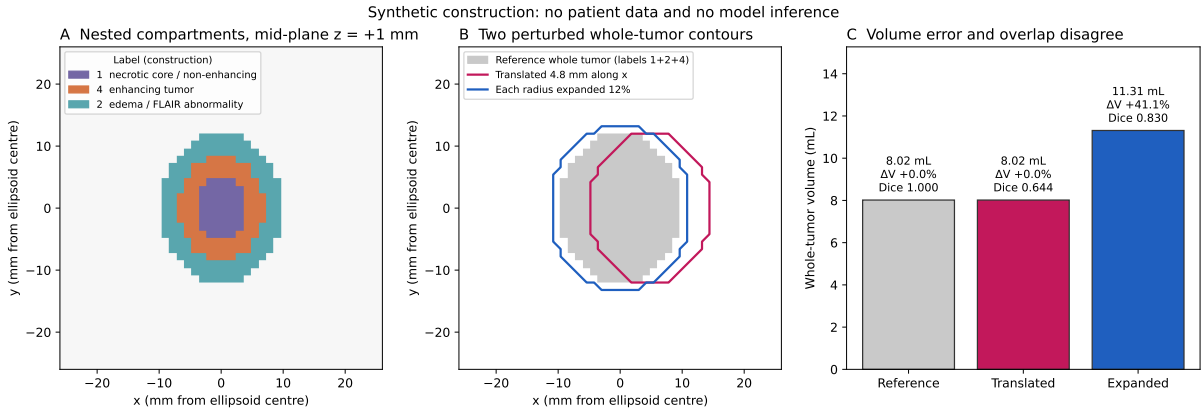


Figure 1: Overlap and volume respond differently to two small segmentation errors (SYNTHETIC). **(A)** Mid-plane of a constructed ellipsoid (semi-axes 10, 12 and 16 mm) with three nested labels: 1, necrotic or non-enhancing (purple); 4, enhancing rim (orange); 2, oedema (teal). **(B)** Reference whole tumour (grey) with the mask translated by 4.8 mm (magenta) and an ellipsoid enlarged by 12% (blue). **(C)** Volume, signed volume error and Dice relative to the reference. The expansion is close to the continuous expectation $1.12^3 - 1 \approx 40.5\%$. Axes are millimetres from the ellipsoid centre, not patient anatomy.

2.2 A published imaging study and its recomputation (notebooks 01 and 09)

Question: What does a published segmentation and anatomical-profiling study measure, and can its cohort agreement be recomputed from released files?
AI contribution: Trained SegResNet segmentation in the published study; research assistance here with attribution, asset pinning, figure reuse and the cohort recomputation.
Investigator choices: The investigator requested this bridge and authorised publishing the cohort inputs as a pinned release asset.
Verification: Published values exported from a checked publication record; SHA-256 of reused figures and the cohort bundle; Dice and volume error recomputed from masks.
Evidence status: PUBLISHED; recomputation without new inference; no clinical-utility result.

Our feasibility study combined multiparametric MRI tumour segmentation with subject-specific anatomical parcellation [2]. It used co-registered T1, contrast-enhanced T1, T2 and FLAIR, the MONAI BraTS three-compartment SegResNet pipeline, and FreeSurfer parcellation in each subject's native space. Intersecting compartments with anatomical regions gives regional volumes and *hit-plots* (Figure 3). Dataset reference masks of UCSF-PDGM provide the segmentation comparison [3]; they are not histological ground truth. Table 2 quotes the reported agreement.

Table 2: Reported median agreement in the published UCSF-PDGM cohort of IDH-wildtype, WHO grade 4 glioblastoma (PUBLISHED) [2]. WT combines NCR, ED and ET; TC combines NCR and ET.

Compartment ($n = 50$)	Dice	HD95 (mm)
Whole tumour (WT)	0.90	4.1
Tumour core (TC)	0.94	2.2
Enhancing tumour (ET)	0.86	2.0

Notebook 01 recomputes this agreement from released files: the dataset reference segmentations and the companion study's SegResNet label maps for the 50 evaluation subjects, published as a pinned release asset with their licences. Over 50 subjects the recomputed median Dice is 0.901 (WT), 0.941 (TC) and 0.862 (ET), consistent with the published values. Volume tells a different story (Figure 2): the median signed volume error is -8.9% (WT), -3.9% (TC) and -14.9% (ET); the model's whole tumour is smaller than the reference in 48 of 50 subjects, and 5 subjects have an enhancing-tumour volume error of at least 40%. A model can agree well in overlap and still be biased in volume, the same lesson as the construction above, now on real masks.

Notebook 09 adds what segmentation cannot see. Glioma cells infiltrate beyond the visible abnormality, so the T2/FLAIR boundary is a threshold on an unobserved density rather than the edge of the tumour [4]; and the cognitive effects of a glioma are distributed over networks rather than confined to the regions it occupies [5, 6]. The study's stated scope is feasibility and infrastructure [2]. Its regional burden supplies anatomical context; relating it to tissue assays, physiology or outcomes is proposed follow-up work.

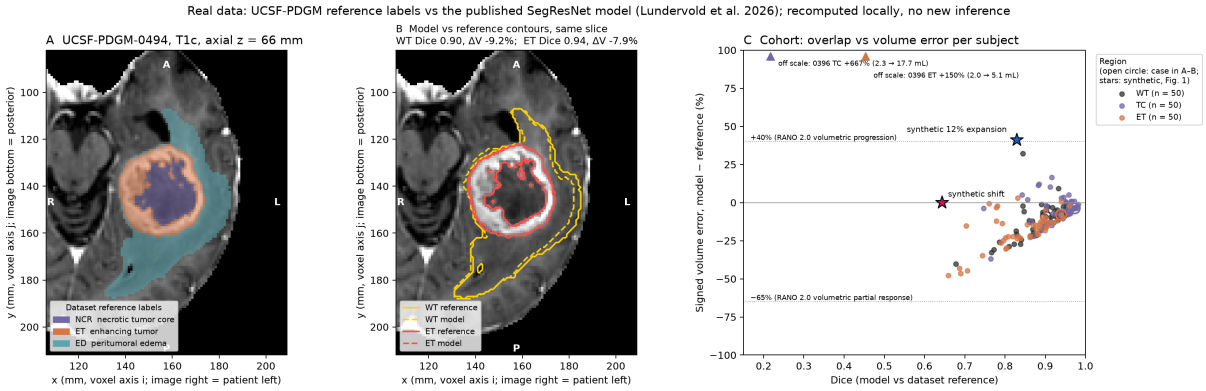


Figure 2: Recomputed agreement between the companion study's SegResNet labels and the UCSF-PDGM reference segmentations for the 50-subject evaluation cohort (PUBLISHED, recomputed). Dice and signed volume error per compartment; no imaging inference was rerun. Data: UCSF-PDGM [3], CC BY 4.0, under the TCIA Data Usage Policy; labels from [2].

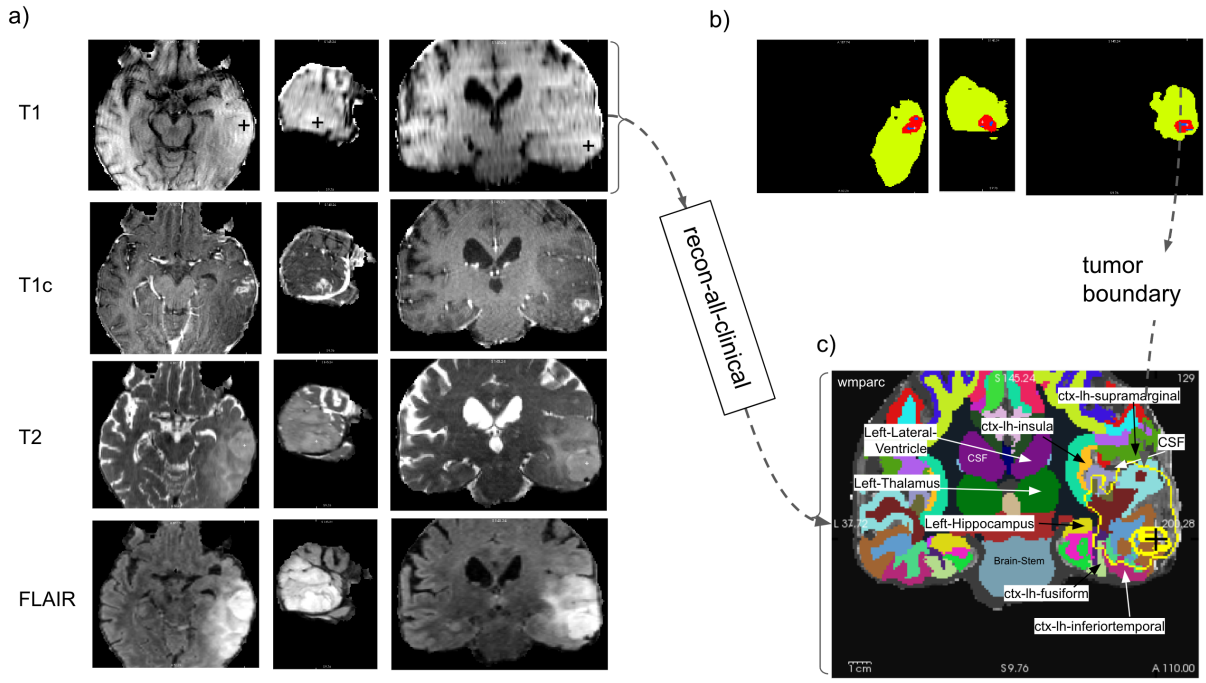


Figure 3: Published imaging illustration (PUBLISHED): multiparametric MRI, dataset-reference compartment masks and subject-specific anatomy for public subject UCSF-PDGM-0085. Original Figure 1 of Lundervold et al. [2], CC BY 4.0, reused unmodified. Anatomical labels locate a possible functional question; they do not measure eloquence.

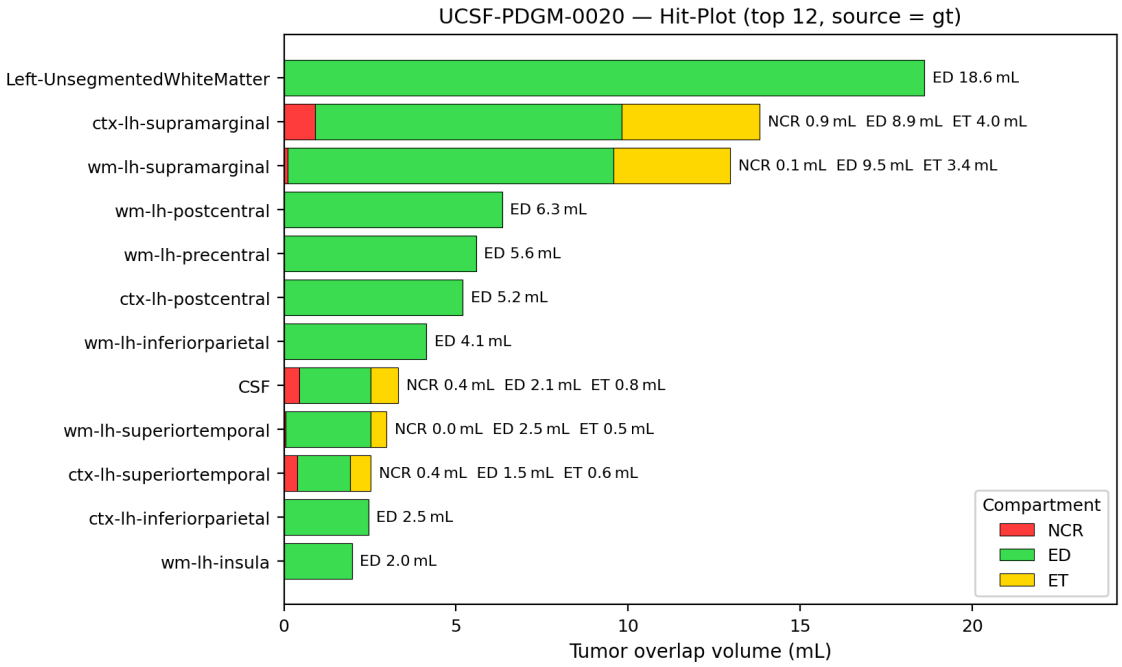


Figure 4: Published regional overlap volumes (mL) of necrosis, oedema and enhancing tumour for public subject UCSF-PDGM-0020 (PUBLISHED). Original Figure 4c of [2], CC BY 4.0. The upstream label `source=gt` identifies dataset reference masks. The twelve regions are the published selection, not a new analysis.

3 One patient's serial scans (notebook 03)

Question: Can a migration–proliferation model, started from the automated labels at one visit, reproduce the automated labels four weeks later, and what does the disagreement teach?
AI contribution: Trained imaging AI (MONAI BraTS SegResNet bundle) produced the labels upstream; research assistance wrote the loader, solver adapter, controls, the biologically informed revision, the fit and figures.
Investigator choices: The investigator chose LUMIERE Patient-048 and authorised publishing the prepared inputs as a pinned release asset under the dataset's non-commercial condition.
Verification: Explicit geometry and label checks, no-change and single-process controls, a discrete balance of the integrated density, and step refinement.
Evidence status: OBSERVATION of automated labels; model runs are SYNTHETIC; the fit is in-sample.

LUMIERE released longitudinal MRI of 91 glioblastoma patients with automated segmentations and expert RANO ratings per visit [7]. Figure 5 shows six visits of Patient-048 on one prepared plane, with labels from a locally run MONAI bundle rather than expert masks. The released ratings record progressive disease at weeks 045 and 049, with a new lesion at week 049. We compare week 045 with week 049, a nominal 28 days, to avoid spanning the resection; the interval still follows treatment.

A baseline that fails. On plane 181 the automated core grew from 840.0 to 1394.0 mm^2. A uniform start at 0.65 inside the core, prescribed coefficients and a threshold of 0.5 give 803.0 mm^2 after 28 days: smaller than the start. Its Dice with the later labels, 0.718, is no better than assuming nothing changed (0.731). The diagnosis is the setup rather than the coefficients: a sharp edge read at one threshold diffuses below that threshold before growth lifts it again.

A biologically informed revision. Following the observation that T1Gd and T2 boundaries correspond to different fractions of the maximum cell density [4], the revision starts from two

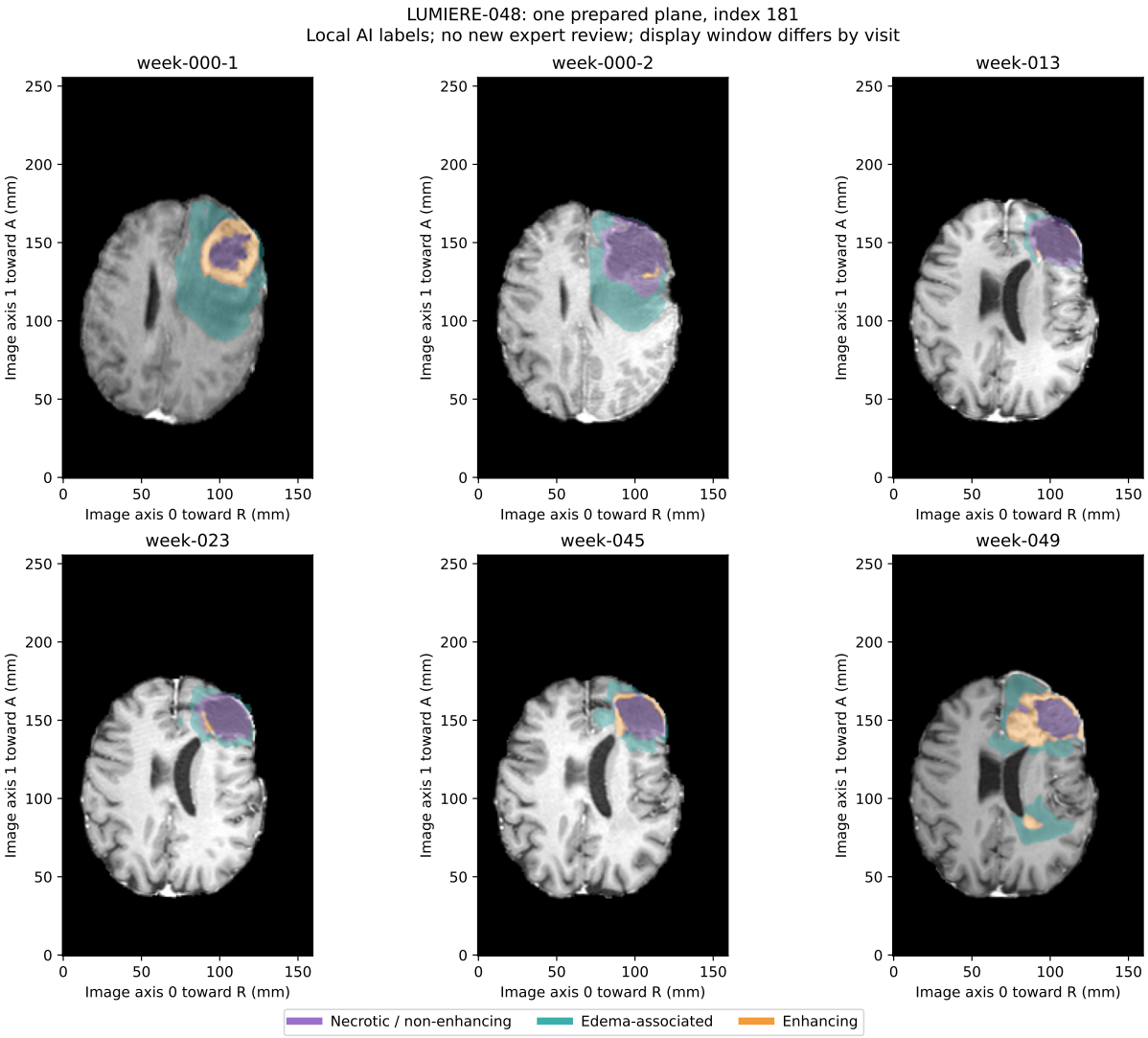


Figure 5: LUMIERE Patient-048, one prepared axial plane over six visits (OBSERVATION). Overlays are automated labels from a local MONAI bundle (necrotic/non-enhancing, oedema-associated, enhancing), not expert segmentations; each CT1 uses its own display window. Visits are categorical positions, not a continuous time axis. Data: LUMIERE [7], used under the authors' non-commercial condition.

levels (core and T2/FLAIR margin), reads the result at two matching thresholds, confines spread to the brain and uses published median coefficients fixed before the comparison [8]. Its local score is 0.784. Part of the later abnormality, 665 mm^2 or 24% of the week-049 lesion on this plane, does not touch the week-045 labels and is out of reach by construction (Figure 6). Fitting the coefficients to week 049 gives $D = 0.251\,\mathrm{mm}^2/\mathrm{day}$ and $\rho = 0.106\,\mathrm{day}^{-1}$ (doubling time 6.6 days) with local score 0.841; the score depends mainly on ρ, so D is barely identified by one interval, and the fit absorbs treatment, scanner and segmentation effects. Notebook 03 repeats the revision in three dimensions with a FreeSurfer tissue map; the distant lesion remains separate in 3D, consistent with the new lesion in the expert rating.

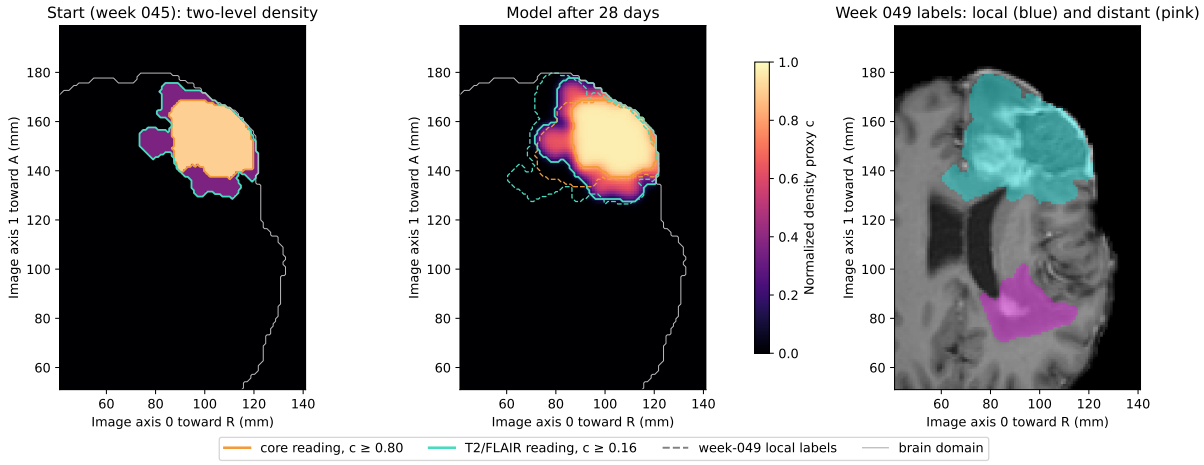


Figure 6: Two-level revision for Patient-048, week 045 to week 049 (SYNTHETIC model on OBSERVATION labels). Left: assumed start density with core and T2/FLAIR readings. Middle: model after 28 nominal days, with the week-049 local labels dashed. Right: week-049 labels split into the component that touches week 045 (blue) and a distant lesion (pink). The density is a normalised proxy, not a measured cell count.

Imaging change and clinical response require different evidence. RANO 2.0 provides criteria for glioma trials that combine imaging with clinical and treatment context [9]; PET RANO 1.0 adds amino-acid PET criteria [10]; RANO/EANO recommendations cover clinical PET use [11]; and the wider RANO effort includes an AI working group [12]. Overlap with a later automated label does not establish a response category or a recurrence prediction.

4 Growth and its environment

The examples in this and the next section resolve different quantities on different timescales. They are not parts of one calibrated multiscale simulator: seconds-scale extracellular potassium, a static potential profile and a day-scale density are not interchangeable.

4.1 A checked proliferation–invasion model (notebook 02)

Question: How do migration, local proliferation and the choice of boundary change a simulated tumour?
AI contribution: Research assistance adapting solver interfaces, controls, the tissue phantom and explanation.
Investigator choices: Tumour dynamics was requested; coefficients, geometry and boundaries are prescribed.
Verification: No-change and single-process controls, conservation without growth, analytic solutions and step constraints.
Evidence status: SYNTHETIC; no fitted patient trajectory.

Serial MRI observes a glioma through boundaries, each delineated with error. Whether such observations could support forecasts of growth has been framed by analogy with numerical weather prediction [13]. Weather forecasting now combines physics-based ensembles with machine-learned models trained on decades of dense observations [14–16]. A glioma is seen in a few scans months apart, its course is altered by treatment, and pseudoprogression can mimic growth [9].

The baseline advances a normalised density c:

$$\frac{\partial c}{\partial t} = \nabla \cdot (D \nabla c) + \rho c(1 - c), \tag{2}$$

the Fisher–Kolmogorov–Petrovsky–Piskunov equation, whose fronts travel at $2\sqrt{D\rho}$ [17–19]. Swanson, Alvord and Murray used the proliferation–invasion form with faster migration in white than grey matter [20, 21]. D is in $\mathrm{mm}^2/\mathrm{day}$, ρ per day and coordinates in millimetres; the boundary has no flux, and steps outside the diffusion stability bound are rejected rather than clipped. From a 7 mm disk of 149 mm^2, the prescribed solution covers 145 mm^2 at threshold 0.5 and 553 mm^2 at 0.05 after 20 days; doubling D gives 109 mm^2 at 0.5 (Figure 7). The same state looks larger or smaller depending on the threshold, and the integrated density can grow while a high-threshold area shrinks.

Notebook 02 then places a left parietal seed on a constructed coronal phantom with grey matter, white matter, CSF and prescribed fibre directions. Faster migration in white matter and along fibres steers the simulated tumour across the corpus callosum. In a recent analysis of recurrence sites in 195 patients with grade 4 glioma, the authors suggest that posterior white matter, including the splenium, may facilitate migration to new lobes [22]. Notebook 11 reuses this phantom and tumour.

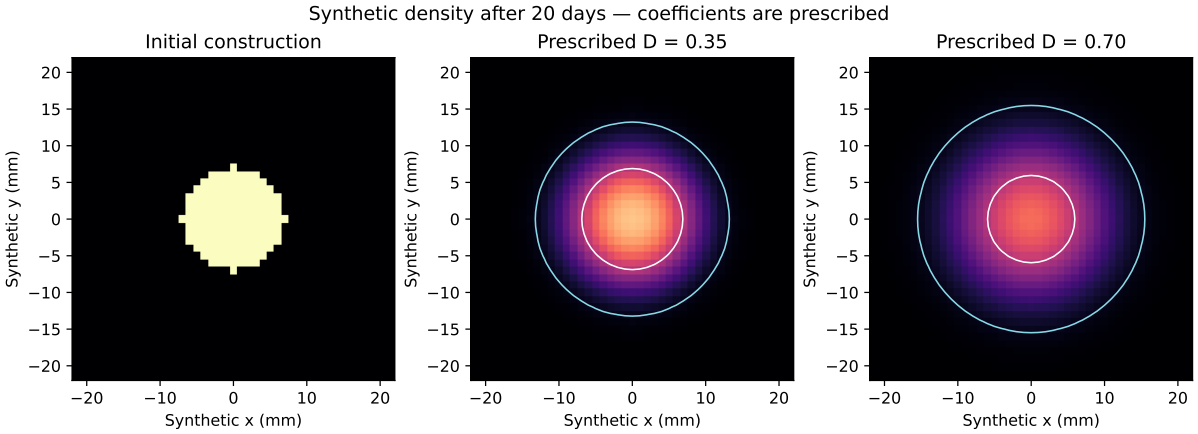


Figure 7: Constructed density and prescribed dynamics after 20 days (SYNTHETIC). Middle and right panels use different prescribed diffusivities; cyan and white contours mark thresholds 0.05 and 0.5. The initial disk is the no-change comparator. Not an MRI prediction or an inferred growth rate.

4.2 A niche signal changes composition at fixed abundance (notebook 05)

Question: Can a glioblastoma change its cell-state composition without growing?
AI contribution: Research assistance implementing the model, comparisons, tests and figures.
Investigator choices: Biological modelling was requested; states, reversibility, cue and rates await scientific review.
Verification: Exact solution, conservation, equilibrium, no-cue limit and partition invariance.
Evidence status: SYNTHETIC; no observed phenotype or fitted niche effect.

Macrophage-associated signalling can drive mesenchymal-like (MES-like) glioblastoma state transitions in experimental systems [23, 24], and hypoxia-associated macrophage niches illustrate

the spatial nature of this ecosystem [25]. Let P aggregate other tumour states and M a MES-like program analogue, both normalised abundances. A prescribed cue $s \in [0, 1]$ gives

$$\frac{dM}{dt} = k_f sP - k_r M, \qquad \frac{dP}{dt} = -\frac{dM}{dt}, \tag{3}$$

which conserves $P + M$ locally. The cue is neither a cell count nor a cytokine concentration, and no rate is inferred from the cited experiments. With $k_f = 0.08$/day and $k_r = 0.02$/day over ten days, the MES-like fraction becomes 27.95% with the cue and 8.19% without it, at identical total abundance (Figure 8). A volume measurement cannot see this change; a molecular or protein assay could, and a perturbation would be needed to test a causal signalling explanation.

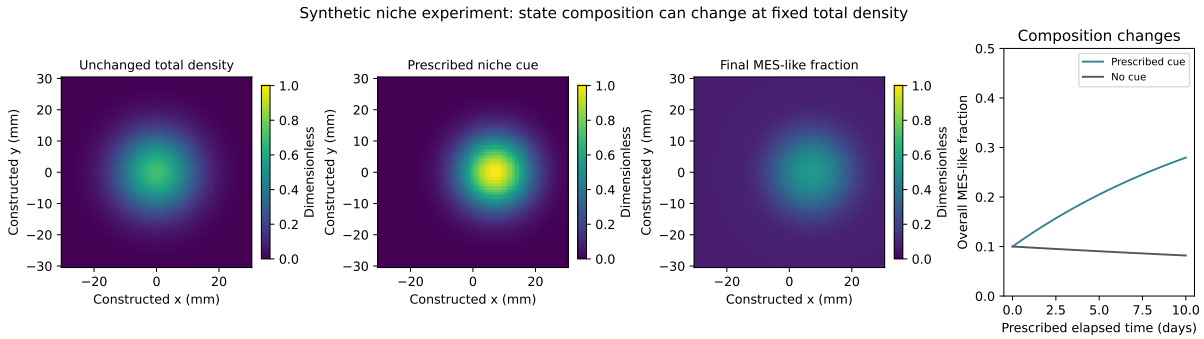


Figure 8: Synthetic niche construction (SYNTHETIC). A prescribed cue changes the distribution of a MES-like program analogue while total normalised abundance is fixed; the line plot compares the cue with its absence. Neither molecular state nor MRI appearance is inferred.

4.3 A neuron–glioma feedback experiment (notebook 06)

Question: Does a glioma feed its own growth, and can a growth curve tell?
AI contribution: Research assistance implementing the model, nested controls and sensitivity.
Investigator choices: Cancer neuroscience was requested; the slow index and coefficients are prescribed choices.
Verification: Uncoupled limit, analytic constant-signal case, conservation, refinement and growth confounding.
Evidence status: SYNTHETIC; no fitted physiology or cognitive endpoint.

Cancer neuroscience connects neurosurgery and tumour biology [26, 27]. Neuron–glioma synapses and electrical integration have experimental support [28, 29]; their plasticity is regulated by BDNF–TrkB signalling [30]; human cortical-tissue experiments link neuronal excitability to glioma proliferation [31]; and tumour functional connectivity is associated with cognition and survival in patients [32]. Tumour microtubes add a cellular network level [33–36], estimated in patient samples through a transcriptional signature [37]; they are far below MRI resolution.

The construction adds a slow, dimensionless growth-signal index a to the density c:

$$\frac{\partial c}{\partial t} = \nabla \cdot (D \nabla c) + \rho(1 + \kappa a)c(1 - c), \tag{4}$$

$$\frac{\partial a}{\partial t} = \frac{b - a}{\tau} + \beta c(1 - a), \tag{5}$$

with prescribed drive b, coupling κ, feedback rate β per day and relaxation time τ in days. The index is not voltage, firing rate, EEG or synapse count. Three nested runs share all other inputs: baseline ($\kappa = \beta = 0$), one-way ($\kappa = 1.5$, $\beta = 0$) and feedback ($\kappa = 1.5$, $\beta = 0.12$/day). The integrated density after 20 days is 67.30, 93.00 and 95.85 mm^2. Feedback adds only 3.06% to the one-way case, so most of the difference from baseline is growth modulation, not feedback;

halving the step changes the result by 0.110%. A constant one-way signal can be absorbed into an effective growth rate $\rho(1 + \kappa b)$, so a density trajectory alone can leave the mechanism undetermined (Figure 9).

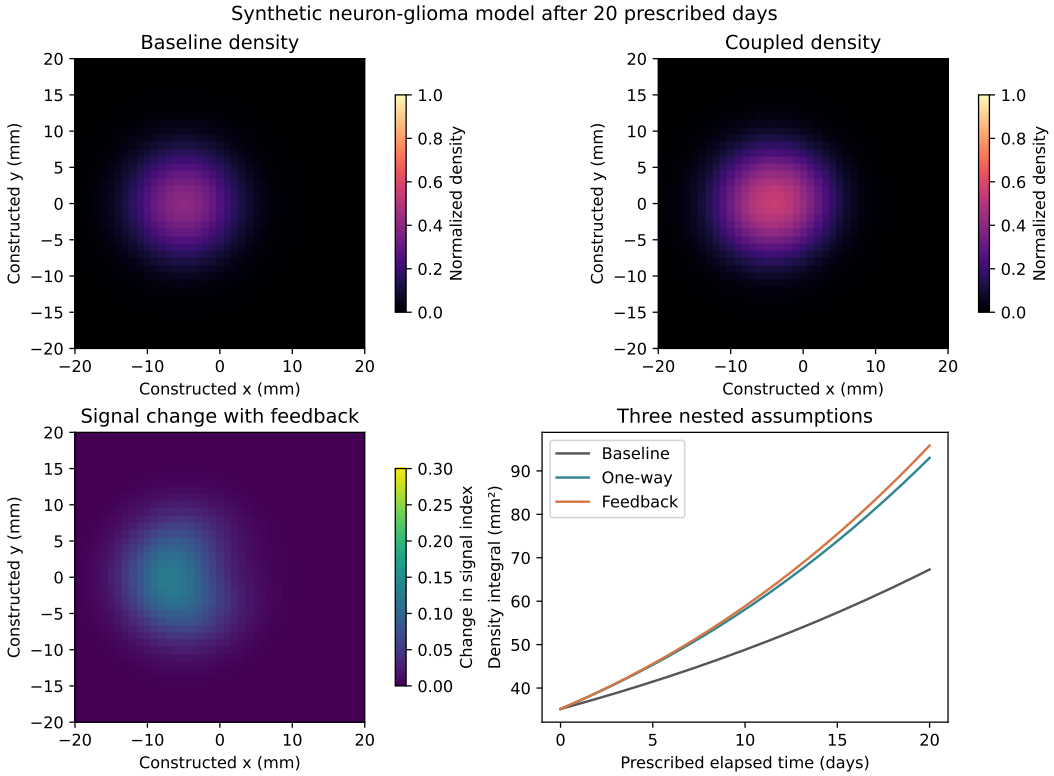


Figure 9: Synthetic neuron–glioma construction (SYNTHETIC). Final baseline and feedback densities on a shared scale, the change of the signal index from its prescribed drive, and traces including the one-way comparator, which separates growth modulation from the additional feedback assumption.

5 Physiology at the glioma–brain interface

5.1 Astrocytic potassium clearance (notebook 07)

Question: When the same activity releases the same potassium, what changes if astrocytic clearance slows?
AI contribution: Codex helped formulate and implement the excess-K and Nernst example and its checks.
Investigator choices: The investigator requested astrocytes; volume, rates, pulses and concentrations are prescribed choices.
Verification: Independent analytic solution, concentration balance, limiting cases, clearance ambiguity and Nernst limits.
Evidence status: SYNTHETIC; scientific review pending.

Astrocytes regulate the extracellular environment; glial-conditional Kir4.1 perturbation links astroglial potassium clearance to short-term synaptic responses [38]. Astrocyte–glioblastoma connections also have metabolic consequences, including GAP43-dependent mitochondrial transfer [39], a process distinct from potassium clearance. Notebook 07 uses one compartment of excess extracellular potassium k (mM) above a 3 mM reference:

$$\frac{dk}{dt} = q(t) - (\lambda_a + \lambda_o)k, \tag{6}$$

with prescribed release q (mM/s) and lumped astrocytic and other clearance rates (s^{-1}). Release is 0.6 mM/s in two four-second pulses; other clearance is 0.1 s^{-1} and astrocytic clearance 0.8, 0.2 or 0 s^{-1}. The ideal K equilibrium potential uses 140 mM intracellular potassium at 37 °C, $E_K = (RT/F)\ln([K]_{out}/[K]_{in})$. Peak excess potassium is 0.649 and 1.524 mM for reference and reduced astrocytic clearance, shifting E_K by up to 5.23 and 10.98 mV (Figure 10). The trajectory depends only on $\lambda_a + \lambda_o$, so a potassium measurement alone cannot attribute clearance to astrocytes within this model; selective perturbation would be needed. E_K is not a simulated membrane voltage or a cognitive endpoint.

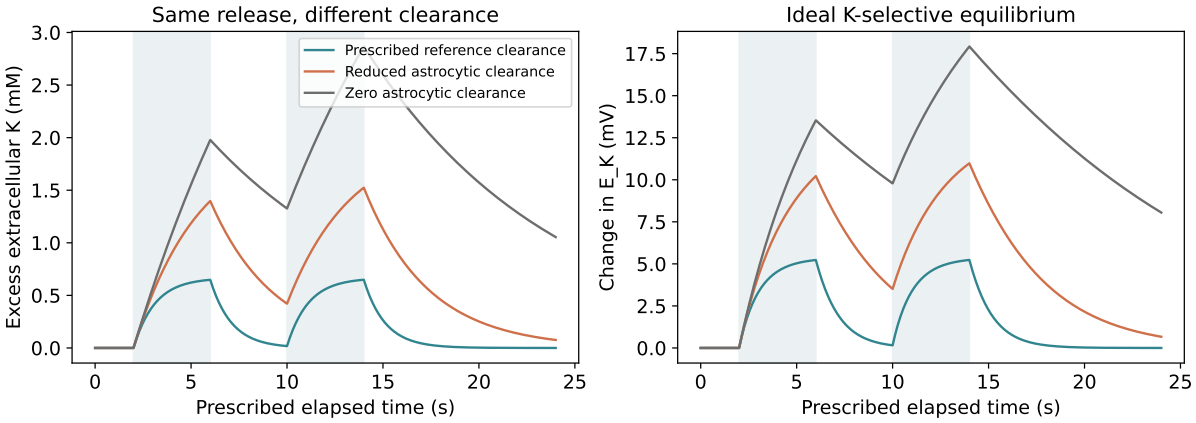


Figure 10: Synthetic potassium-clearance construction (SYNTHETIC). Release intervals are shaded; all scenarios share release and other clearance. The right panel shows the change in ideal K equilibrium potential, not membrane potential, firing or cognition. No glioma or astrocyte observation set the coefficients.

5.2 Potential is not field (notebook 08)

Question: Does a recorded potential identify a spatial field or a cognitive mechanism?
AI contribution: Research assistance with the spatial derivative, controlled examples and evidence map.
Investigator choices: The investigator requested LFPs and cognition; profiles and sampling are prescribed teaching choices.
Verification: Analytic derivative, sign, V/m conversion, offset invariance and primary-source scope.
Evidence status: SYNTHETIC fields; cited experiments and HYPOTHESIS kept separate.

Local field potentials record extracellular potential relative to a reference; synaptic currents, cellular geometry, synchrony and volume conduction contribute [40]. A potential in volts and its gradient in volts per metre are different quantities, $\mathbf{E} = -\nabla\phi$. Weak extracellular fields can influence membrane potential and spike timing in cortical slices [41]. Notebook 08 prescribes two static potentials of equal 100 μV amplitude with wavelengths 10 and 2.5 mm. Their peak fields are 0.063 and 0.251 V/m (Figure 11): equal amplitudes, or a single voltage trace, do not determine the field. A real estimate needs spatial sampling, reference and noise treatment and a source model.

Two theoretical proposals motivate questions here without being tested (HYPOTHESIS). Pinotsis, Fridman and Miller propose that electric fields could influence cytoskeletal organisation [42], and prefrontal recordings show field-to-neuron influences stronger than the reverse [43]; both concern neurons, not glioma cells. Disrupting connexin-43–microtubule interactions reduced glioblastoma stem-like cell survival and xenograft growth [44], which establishes cytoskeletal relevance but not a field-driven mechanism. Intracellular microtubules and tumour microtubes are different structures.

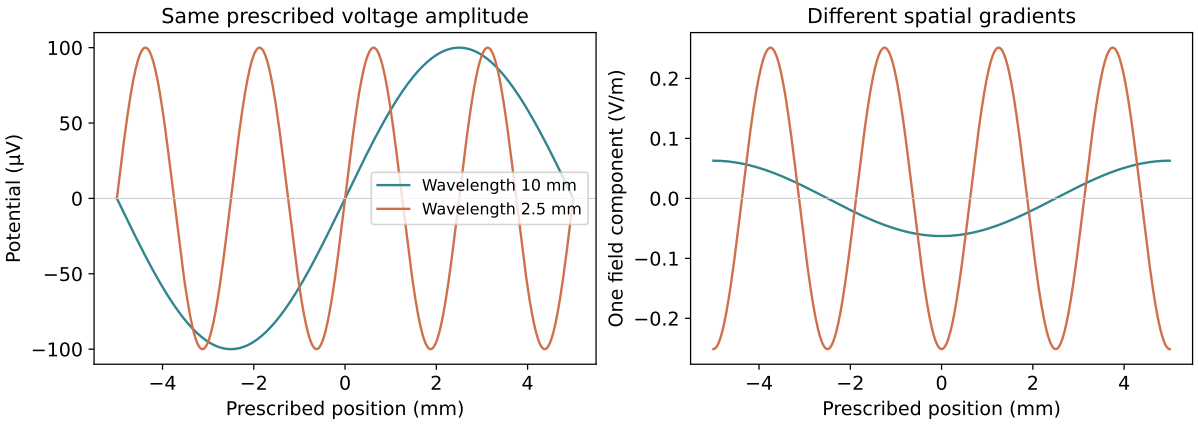


Figure 11: Synthetic spatial potentials and their gradients (SYNTHETIC). Prescribed curves, not recorded LFPs; the computed field is one spatial component, and ephaptic responses, microtubules, growth and cognition are not computed.

6 Networks and cognition

6.1 A tumour in a synthetic network (notebook 11)

Question: When a glioma infiltrates white matter, and part of it is resected, which connections are lost, and how does model functional connectivity change?
AI contribution: Research assistance wrote the network construction, damage rule, linear model, controls, figures, explorer and tests.
Investigator choices: The investigator asked for a link between tumour growth (notebook 02) and cognition (notebook 10); the weights, damage rule and cavity threshold are prescribed.
Verification: Exact stationary covariance checked against simulation, the $g = 0$ limit and step refinement.
Evidence status: SYNTHETIC; nothing fitted to a patient.

Deficits are often discussed as interrupted connections rather than only as occupied cortex [45], and structural connectivity partly predicts resting-state functional connectivity [46]. In stroke, measures of structural disconnection explained functional-connectivity disruption better than measures of focal cortical damage [47]; in glioma surgery, serial mappings suggest that cortex can reorganise extensively while white-matter connectivity limits that plasticity [48]. Notebook 11 places fourteen regions on the coronal phantom of notebook 02, draws tracts as shortest white-matter paths, weights them by an exponential distance rule and lowers each tract's integrity where the tumour density crosses it. Model activity follows $d\mathbf{x}/dt = -\mathbf{x} + gW\mathbf{x} + \sigma\boldsymbol{\xi}$, whose stationary correlation is exact for symmetric W; it is not a BOLD signal. At day 180 the tissue-guided tumour leaves 52.0% of the healthy interhemispheric weight and 88.4% of global efficiency; with the corpus callosum assumed uninvolved, 59.7% would remain. Mean interhemispheric model correlation falls from 0.195 to 0.056. A resection at day 120 removes 763 mm^2 of infiltrated white matter and lowers the strength of cortical regions it spares (Figure 12). Under these rules disconnection follows from the construction, not from biology, and the mapping of regions to cognitive measures in the notebook is unsourced and labelled as such.

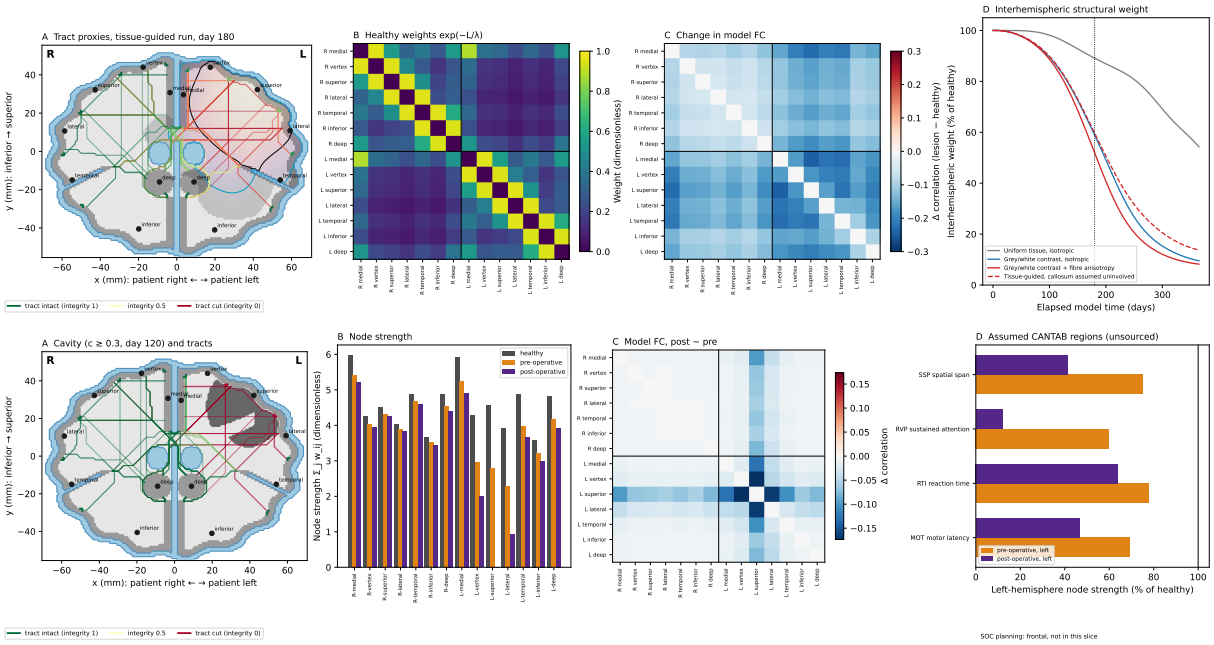


Figure 12: A tissue-guided tumour and a constructed resection on a synthetic network (SYNTHETIC). Top: tumour density proxy at day 180 with tracts coloured by assumed integrity, healthy weights, change in model functional connectivity, and interhemispheric weight over the year for the growth scenarios of notebook 02. Bottom: resection cavity, node strength before and after resection, change in model functional connectivity, and left-hemisphere strength over regions assigned by an unsourced table to four CANTAB measures. Radiological display.

6.2 One patient's connectome before and after surgery (notebook 12)

Question: Does the chain of notebook 11 hold on a real patient: do the changes predicted by a virtual resection resemble what was measured after surgery?
AI contribution: Research assistance inspected the upstream derivatives, proposed rebuilding the structural matrices, and wrote the analyses, figures, explorer, tests and text.
Investigator choices: The investigator authorised tracking the small CC0-derived matrices and publishing them with the upstream quality control stated as unreviewed.
Verification: Shuffled-region and random-placement nulls, the agreement obtained from the measured post-operative structure, provenance hashes and tests.
Evidence status: OBSERVATION, one patient, descriptive; upstream QC not reviewed.

sub-PAT16 is a participant in the open Brain Tumor Connectomics data, with pre-operative and roughly six-month post-operative diffusion and resting-state functional MRI [49–51]. An upstream pipeline, not in the repository, produced SIFT2-weighted streamline counts [52] between 82 Desikan–Killiany regions and a tumour mask from a trained segmentation model; the repository tracks only the derived matrices. A virtual resection removes every streamline that passes through the mask, as Aerts and colleagues did with The Virtual Brain in this cohort [53, 54]. The mask measures 53.3 mL and carries 10.6% of the pre-operative structural weight. Agreement between model and measured functional connectivity is weak: $r = 0.113$ before surgery, at the upper edge of a shuffled-region null (97.5th percentile 0.113), and $r = 0.297$ after. The virtual resection's predicted functional change correlates with the measured change at $r = -0.145$, against a median of -0.090 for random placements of the same lesion and $r = 0.044$ when the measured post-operative structure is used instead (Figure 13). Homotopic pairs have mean functional connectivity 0.68 against 0.29 for other interhemispheric pairs, while the model gives them 0.04: interhemispheric structure carries only 1.9% of the pre-operative weight and 9.2% of the post-operative weight, a difference more likely due to processing than to surgery. In this patient, the model adds nothing beyond the direct correlation with structural weight, and the

measured structural change does not follow the virtual resection.

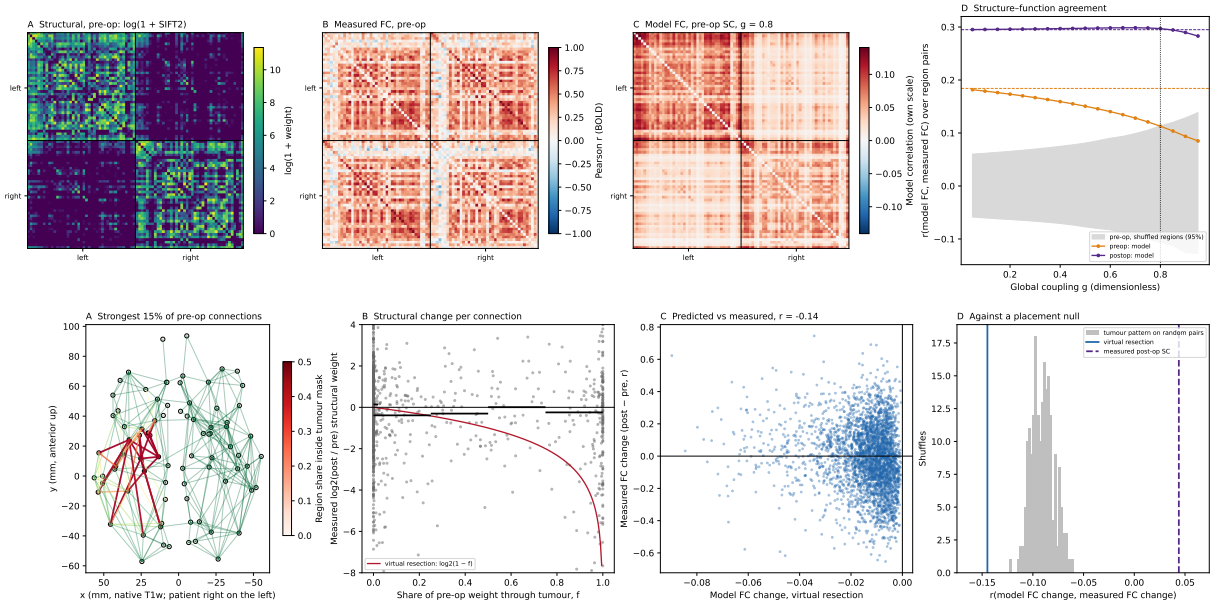


Figure 13: Structure, function and a virtual resection in one patient (OBSERVATION; upstream QC not reviewed). Top: pre-operative structural connectivity $\log(1 + S)$, measured and model functional connectivity, and model–measured agreement against coupling g before and after surgery with a shuffled-region null. Bottom: region centroids coloured by tumour share with the connections that run through the tumour; measured structural change against the share of weight through the tumour, with the virtual-resection expectation; predicted against measured functional change; and that agreement against random placements. Data: Brain Tumor Connectomics, CC0 [50, 51].

Notebook 12 also offers an explorer of resection size and location by a geometric proxy: a sphere removes the streamlines that pass through it. A sphere of the mask's volume (radius 23.3 mm) removes 11.3% of the structural weight, against 10.6% measured through the mask, and recovers 76% of the strongly affected connections with 71% precision. The 20 mm example in Figure 14 removes 8.5%. The explorer is didactic, not a planning tool: the patient's actual cavity is not used, and the cognitive groupings are an assumed table. Recent reviews associate connectome- and tractography-guided resection with tract preservation and postoperative cognition [55], and network accounts of glioma reach from tumour biology to symptoms [6, 56].

6.3 Cognition before and after surgery (notebook 10)

Question: What did patients and their partners score before surgery and at follow-up, how far can one patient's change be distinguished from measurement error and practice, and what would prediction require?
AI contribution: Research assistance pinned the participant tables, wrote the group summaries, change indices, bootstrap and leave-one-out forecasts, and transcribed and audited an AI-generated slide.
Investigator choices: The investigator supplied the slide and chose sub-PAT16 as the case; demographic and mood variables are not shown under a privacy decision.
Verification: SHA-256 pins, agreement of three sources at released precision, and change indices against the retested controls.
Evidence status: OBSERVATION; change indices exploratory; no causal surgical effect or trained predictor.

The Brain Tumor Connectomics cohort released Cambridge Neuropsychological Test Automated Battery (CANTAB) [57] scores before surgery and at follow-up under CC0 [58, 59]; a behavioural release accompanies the authors' study of emotion regulation and cognition [60, 61]. Controls are the patients' partners. CANTAB scores before surgery exist for 11 controls, 14

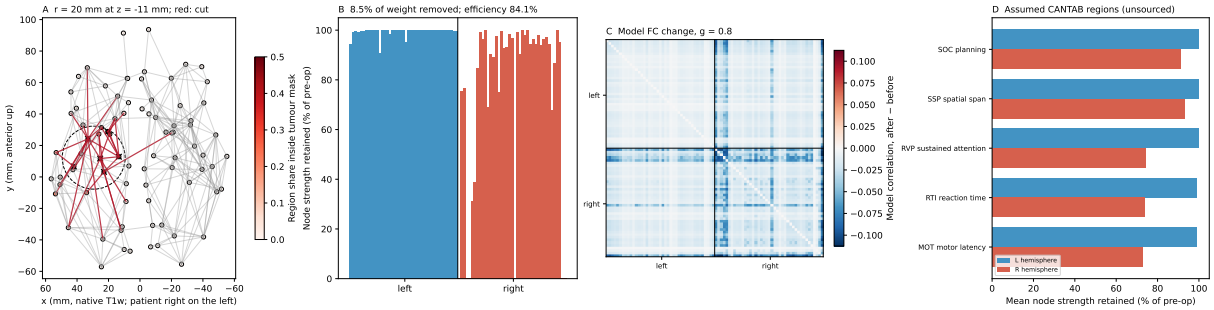


Figure 14: A spherical virtual resection of radius 20 mm at the tumour-weighted centre, by geometric proxy (OBSERVATION of the pre-operative matrices; lesion SYNTHETIC). Region centroids and the strong connections the sphere cuts, retained node strength per region, model functional change, and mean retained strength over assumed, unsourced region sets for four CANTAB measures. No prediction is made for this patient.

meningioma and 10 glioma participants; follow-up rows exist for 10, 12 and 7 of them. Each measure keeps its own unit and direction (Figure 15); the groups overlap widely, and unpaired medians at the two sessions describe different people.

For sub-PAT16, the patient of Section 6.2, three public sources agree on every released score. An AI-generated summary slide of the same scores, supplied by the investigator, did not: it showed values impossible on the named scales (a sensitivity index above 1, more planning problems than the test contains), percentages that contradicted its own bars, and "significant" changes without a test. The notebook keeps the slide unmodified as an example for auditing next to a reconstruction from the released scores. Reliable-change indices [62–64], computed against the ten retested controls, put none of the case's changes beyond the conventional threshold in the main analysis; the closest, a faster five-choice reaction time, crossed it when one control was left out. In leave-one-out forecasts of follow-up scores, carrying the baseline score forward was hard to beat. Any model of cognition after surgery should be compared with that rule on the same held-out patients, with sample sizes justified in advance [65] and reporting according to TRIPOD+AI [66].

6.4 From networks to cognition: hypotheses and a proposed study

Cognition belongs in the research agenda as an explicitly assessed outcome, not a label on a simulated quantity. Miller, Brincat and Roy propose a hybrid synapse/wave account in which rhythmic fields organise context-dependent activity [67]. Asking how glioma-related perturbations of such dynamics might affect cognition beyond anatomical burden is our HYPOTHESIS; testing it needs task-related EEG, MEG or intracranial recordings, not BOLD or behavioural scores alone.

We propose (PROPOSED) asking whether pre-operative network and dynamics information adds predictive value beyond baseline cognition, clinical information and reviewed anatomy, with follow-up spatial span as a candidate primary endpoint. A persistence rule is the first comparator, followed by clinical, anatomical and then a few network or independently fitted dynamics features, all in the same compact regularised model family. Predictors must exist before surgery; preprocessing belongs inside training folds; all visits of one person stay together; and errors are reported in native units. Earlier virtual-resection work showed variable agreement with post-operative functional connectivity [54], and the single patient of Section 6.2 agrees no better. A larger, linked glioma cohort is required.

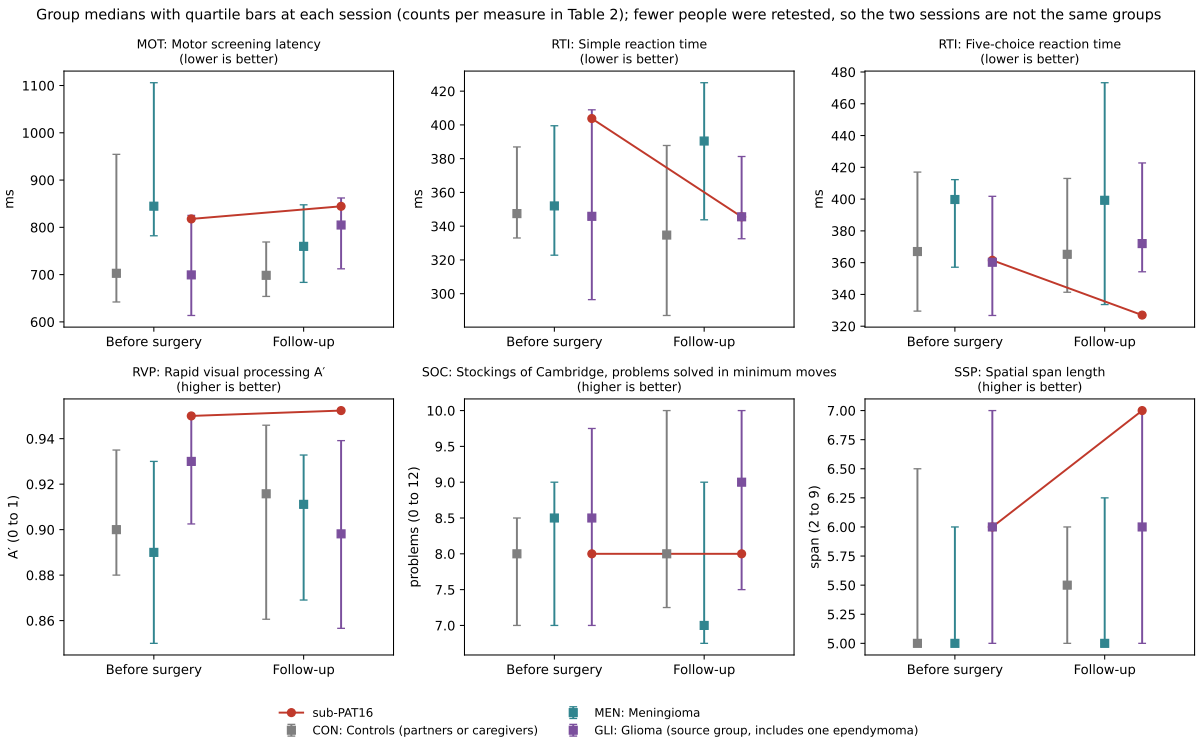


Figure 15: CANTAB group medians with quartile bars before surgery and at follow-up, with sub-PAT16 in red (OBSERVATION). Each panel keeps its native unit and states whether higher or lower is better. Fewer people were retested, so the two sessions are not the same groups. Data: OpenNeuro ds001226 v5.0.1 and ds002080 v4.0.1, CC0 [58, 59].

7 Auditing AI-assisted research (notebook 04)

Question: When a generative assistant helped write scientific code, which evidence lets another researcher judge that contribution, and which checks could have caught a mistake?
AI contribution: Research assistance wrote the records, the record validator and the constructed faulty solvers.
Investigator choices: The investigator requested explicit roles; human scientific approval remains a separate decision.
Verification: Each faulty solver must fail exactly the checks expected of it; the notebook asserts every verdict.
Evidence status: Saved retrospective records and SYNTHETIC teaching examples; no transcript, measured benefit or human review.

Four questions about assisted work are easily merged into one (Table 3). A record of what an assistant contributed answers the first; independent checks answer the second; neither answers the last two. The saved records in the repository are retrospective summaries, all marked as pending human review, and none captured the model snapshot.

A check is evidence only if it could have failed. Notebook 04 derives three checks from the potassium equation alone (agreement with the closed-form solution, a balance of released, retained and cleared potassium, and invariance when the two clearance rates are swapped) and applies them to four candidate solvers, two of them deliberately faulty (Table 4). Each faulty candidate fails at least one check and passes another; no single check catches all of them, and the correct solver passes all three at rounding level. Two faults are invisible on a plot. The tolerance decides whether forward Euler counts as an error, and inputs can blind a check: with no other clearance, the bug that omits it passes the analytic and balance checks.

A record validator checks form, not truth. It refuses a missing date, a template placeholder, an empty list of checks, or a decision of *accepted* without a named reviewer; it cannot tell whether

Table 3: Four questions about AI-assisted research code, the evidence that could answer each and their status for the potassium example of Section 5.1.

Question	Evidence that can answer it	Status
Completeness	A structured record of task, tool, inputs, contribution, checks, decision and open questions	Record exists and validates
Correctness	Independent checks that would fail if the code were wrong	Checks pass
Scientific quality	Review by someone qualified to judge the model and its interpretation	Pending human review
Benefit	Comparison of assisted and unassisted work on matched tasks	Not performed

Table 4: Three checks applied to four candidate solvers of Equation (6) (SYNTHETIC). The two bugs are constructed for teaching, not taken from the development history. Verdicts at the notebook's tolerances; the notebook asserts each one.

Candidate	What it does	Analytic	Balance	Swap
Package	Exact update over each step	pass	pass	pass
Forward Euler	$k_{n+1} = k_n + \Delta t\,(q - \lambda k_n)$	fail	pass	pass
Bug: omits other clearance	Uses only λ_a	fail	pass	fail
Bug: clearance booked from step start	Correct k, cleared amounts from k at the start of each step	pass	fail	pass

the checks were run, whether they are adequate, or whether a named reviewer reviewed anything.

Benefit is a separate question. A proposed pilot (PROPOSED) separates productivity, scientific quality and clinical relevance: two investigators each complete six matched task pairs, one version of each with a specified generative assistant and the other without, in counterbalanced order. The primary measure is quality-qualified completion time with a 60-minute cap; independent reviewers blinded to condition rate units, correctness, controls, provenance and evidence boundaries. No session has been run, and the design cannot establish general superiority.

8 Discussion

The contribution of these examples is a chain another researcher can inspect: question, specification, contribution, check, output and the boundary of what the output supports. The notebooks call one package, the paper's numbers and figures come from the same functions, and the release assets that hold licensed inputs are pinned by hash.

Three real-data examples show how narrow the supported conclusions are. The cohort recomputation confirms the published overlap and adds a systematic volume bias that overlap alone hides. The LUMIERE case shows that a model can be revised for stated biological reasons and still not identify its coefficients from one interval, while part of the later lesion lies outside any local model. The sub-PAT16 connectome shows that the simplest network chain, from tumour to lost connections to model functional connectivity, does not describe one patient's measurements, with the upstream quality control still unreviewed. The CANTAB example shows that one patient's change can hinge on the choice of index and on one reference person, and that an AI-generated summary of correct numbers can still be wrong.

The constructions serve a different purpose. They make assumptions visible and testable: a threshold that changes apparent size, a composition that changes without growth, a feedback that a growth curve cannot distinguish from modulation, a clearance that a potassium trace cannot attribute, and a potential that does not determine a field. None is calibrated, and none

predicts cognition.

Notebook 13 places these examples in a research workflow of nine stages: idea, literature review, research questions, study design, applications (ethics approval and financing), implementation, data analysis, reporting and further studies. It distinguishes three modes of assistance: support for one bounded request, collaboration over a dialogue, and agents that carry out several steps with tools between human checks. Most examples in this paper belong to implementation, analysis and reporting. Checks matter at the earlier stages too. In this project, a first author named in a request was wrong, and a dataset portal's CC0 label differed from the non-commercial condition stated by the dataset's authors; both were found by comparison with the source. In one study, 55% and 18% of the citations generated by two ChatGPT versions were fabricated [68]; data leakage has been reported in 294 machine-learning-based papers across 17 fields [69]; and AI tools may create illusions of understanding, in which researchers believe they understand more than they do [70]. Applications for ethics approval and funding, where confidentiality constrains AI use most, are discussed in the notebook but not exemplified here. The notebook adds no result (PUBLISHED; PROPOSED).

The limitations are substantial. Synthetic shapes omit biological, acquisition and treatment variability. The growth models have prescribed or in-sample coefficients and no treatment term. The network model is linear and has no haemodynamics. The real-data examples are one patient each, or a small mixed cohort. The analog-cognition and cytoelectric interpretations remain hypotheses. The records of AI assistance are retrospective summaries; none of these artifacts measures the comparative benefit of AI assistance, and no clinical utility is claimed.

9 Conclusion

Glioma research spans images, cells, circuits, networks and cognition, and each level answers a different question. Twelve executable notebooks keep their units, assumptions, evidence labels and limits explicit, from segmentation agreement to one patient's connectome and cognitive scores; a thirteenth places AI assistance in each stage of a research workflow. AI supports these artifacts in three distinct roles. Records and checks that could fail make an assisted contribution inspectable; scientific review, comparative benefit and clinical utility remain to be established.

Declarations

AI assistance. Generative assistants helped design the repository and draft the code, notebooks, tests, documentation and this manuscript: Codex desktop in early October 2026, Cursor agents from 10 October 2026 (reported as Claude Opus 5.5 on 15 and 19 October), and Antigravity for the Colab setup on 14 October. The assistance records in `docs/ai-records/` and `docs/ai-use/` are retrospective summaries; model snapshots and sampling settings were mostly not captured. Rebuilding the examples and this paper calls no language model. Human scientific review remains pending.

Code and data. Code, notebooks and this paper are at `https://github.com/arvidl/AI-assisted-glioma-research` under the MIT licence, with selected helpers adapted from a predecessor repository at commit `f8a37993d11f595135f407638750442041422391`. Earlier development took place in a private repository that is now archived. Licensed inputs are not covered by the MIT licence and are distributed as pinned release assets or fetched from their sources: UCSF-PDGM labels (CC BY 4.0, TCIA Data Usage Policy), LUMIERE Patient-048 prepared inputs (non-commercial condition of the dataset authors), the published figures (CC BY 4.0), the Brain Tumor Connectomics tables and derived sub-PAT16 matrices (CC0) and the OSF behavioural release (CC BY 4.0). Images, tractograms and demographics are not in Git.

Authorship, funding and competing interests. The author and affiliation follow the investigator's instructions. Collaborator contributions, funding, competing interests and target journal remain to be completed. No co-author approval or clinical evaluation is asserted.

A Reproduction

Install the locked environment and run:

```
make setup
make check
make notebooks
make paper
```

`make paper` builds every figure and number in this paper from the package (`make paper-assets`) and then compiles the LaTeX source. The real-data figures need explicit fetch steps first: `make btc-fetch`, `make published-figures-fetch`, `make hpgs-bundle-fetch` and `make lumiere-bundle-fetch`, as described in `docs/data.md`. Missing inputs produce a notice in place of the figure; nothing is substituted. Each notebook opens in Google Colab from its badge; notebook 00 is a quickstart. Executed copies of notebooks 01–13 are tracked in `notebooks/executed/` with their source licences, and `docs/validation.md` records the platforms on which the checks actually ran.

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